

O1Lite ROS driver简单使用说明

环境配置

1.O1Lite/O2S设备直连接电脑或者开发板

设置电脑和模组统一网段，模组ip：192.168.1.251

电脑设置除了192.168.1.x(不能和模组同ip)



1.ROS driver准备

1.创建一个src目录

2.src目录解压我们压缩包到当前目录，命令示例：tar -xvf

ros_driver_v0.5.0_release_20260311.tar.xz

目录结构如下

```
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ls
check_usac  check_usac.cpp  src
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$

xxx@xxx-MS-7E34:~/work/odin2/ros_fold/src$ ls
ros_driver
xxx@xxx-MS-7E34:~/work/odin2/ros_fold/src$
```

3.编译,返回src目录上一级，执行命令

ROS2 (./src/ros_driver/script/build_ros2.sh)

ROS1 (./src/ros_driver/script/build_ros1.sh)

```
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ./src/ros_driver/
config/ docs/ format.sh genpubpkg.sh .git/ include/ launch/ module/ rviz/ script/ src/ srv/
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ./src/ros_driver/script/build_ros2.sh
```

4.配置ros编译环境

ROS2(source install/setup.bash)

ROS1(source devel/setup.bash)

```
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ source install/setup.bash
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ls
build check_usac check_usac.cpp install log src
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$
```

5.运行

ROS2(ros2 launch odin_ros_driver_rev1 driver.launch.py)

ROS1(roslaunch odin_ros_driver_rev1 driver.launch)

###进行完以上步骤后rviz会自动显示图

方式一.Rqt界面操作方式(推荐)

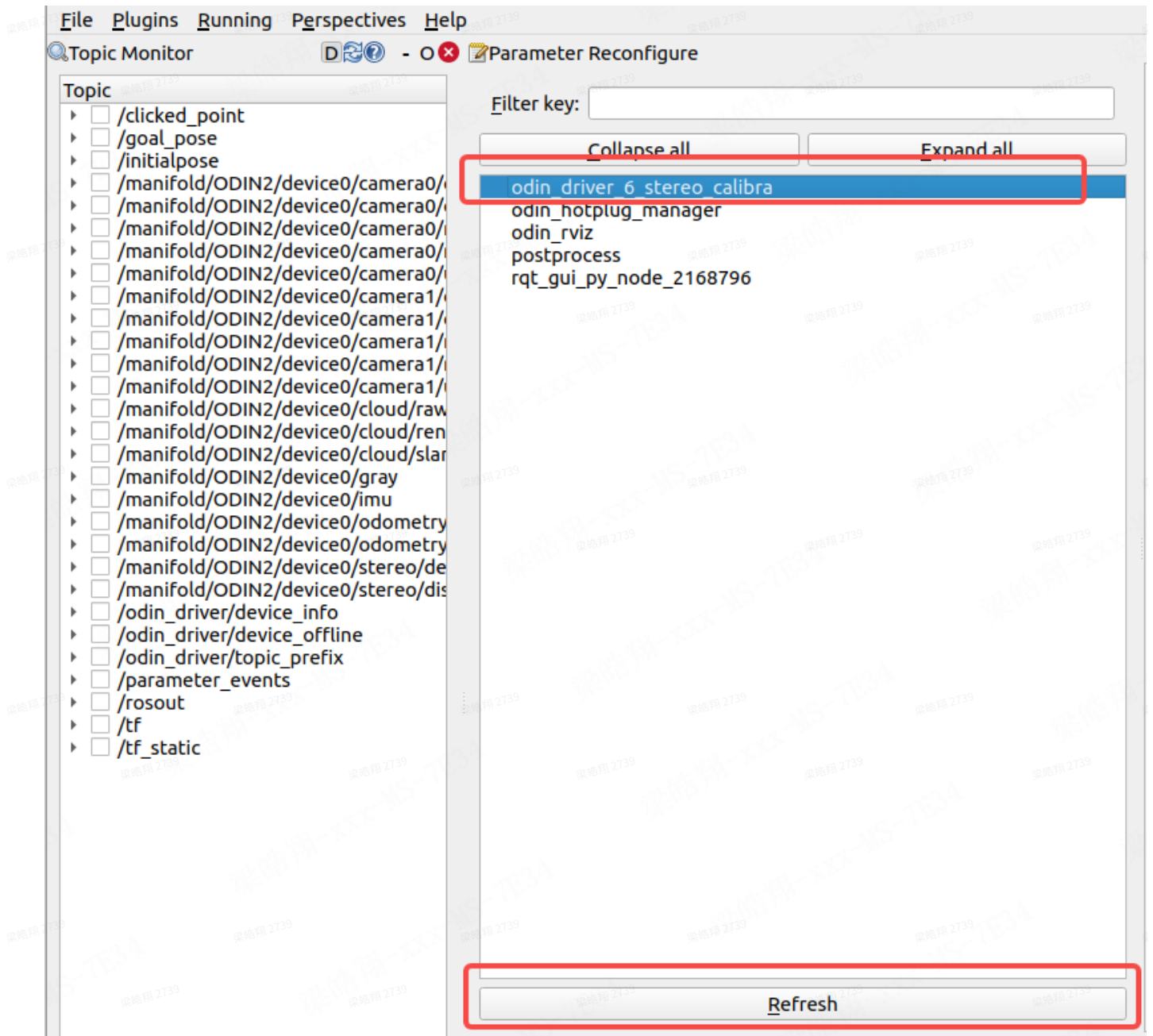
操作步骤

一.命令行输入rqt(ROS1/ROS2操作相同)

```
^C[INFO] [1773304220.528927060] [rclcpp]: signal_handler(signum=2)
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ rqt
```

二.点击Plugins->Configuration->Dynamic Reconfigure

三.点击Refresh后，选择对应模组的节点odin_driver_xxxx(xxx为实际模组SN)



强弱光模式切换

设备的激光模组现支持两种工作模式：

1. High peak模式，建议户外场景使用
2. Low peak模式，建议室内场景使用，近距离点云质量有较大优化

目前支持通过rqt进行动态切换

auto_discover	☐
command_timeout_ms	1000
confidence_threshold	0 <input type="range"/> 255 7
device_command_port	60001
device_ip	192.168.2.251
device_sn	
discovery_timeout_ms	2000
file_data_port	60000
gray_image_frame_id	odom
host_command_port	60002
host_ip	192.168.2.10
image2_frame_id	odom
image_frame_id	odom
imu_data_port	59003
imu_frame_id	odom
jpeg_data_port	59002
jpeg_data_port2	59005
odom_angular_velocity_scale	1e-06
odom_child_frame_id	base_link
odom_data_port	59004
odom_frame_id	odom
odom_linear_velocity_scale	1e-06
odom_orientation_scale	1e-06
odom_position_scale	1e-06
operating_mode	normal
point_data_port	59000
qos_overrides./parameter_events.publisher.depth	1000
qos_overrides./parameter_events.publisher.durability	volatile
qos_overrides./parameter_events.publisher.history	keep_last
qos_overrides./parameter_events.publisher.reliability	reliable
raw_frame_id	odom
sensor_mode	1 <input type="range"/> 2 1
slam_data_port	59001
slam_frame_id	odom
use_sim_time	☐

其中 sensor_mode 1 对应High peak模式，sensor_mode 2 对应low peak模式

点云置信度过滤

点云数据点的confidence的值小于confidence threshold的时候，对应点云就会被设置为xyz = 0的点。

方式二.终端ros服务方式(ROS1/ROS2 service)

操作步骤

ROS2

一.终端输入 ros2 service list /查看服务列表

```
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ros2 service list
/manifold/ODIN2/device0/get_calibration
/manifold/ODIN2/device0/set_sensor_mode
/odin_driver_6_stereo_calibra/describe_parameters
/odin_driver_6_stereo_calibra/get_parameter_types
/odin_driver_6_stereo_calibra/get_parameters
/odin_driver_6_stereo_calibra/list_parameters
/odin_driver_6_stereo_calibra/set_parameters
/odin_driver_6_stereo_calibra/set_parameters_atomically
/odin_driver_P020100011/describe_parameters
/odin_driver_P020100011/get_parameter_types
/odin_driver_P020100011/get_parameters
/odin_driver_P020100011/list_parameters
/odin_driver_P020100011/set_parameters
/odin_driver_P020100011/set_parameters_atomically
/odin_hotplug_manager/describe_parameters
/odin_hotplug_manager/get_parameter_types
/odin_hotplug_manager/get_parameters
/odin_hotplug_manager/list_parameters
/odin_hotplug_manager/set_parameters
/odin_hotplug_manager/set_parameters_atomically
/odin_rviz/describe_parameters
/odin_rviz/get_parameter_types
/odin_rviz/get_parameters
/odin_rviz/list_parameters
/odin_rviz/set_parameters
/odin_rviz/set_parameters_atomically
/postprocess/describe_parameters
/postprocess/get_parameter_types
/postprocess/get_parameters
/postprocess/list_parameters
/postprocess/set_parameters
/postprocess/set_parameters_atomically
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$
```

二.终端输入source install/setup.bash(谨记)

三.服务命令例子:ros2 service call /manifold/ODIN2/device0/set_sensor_mode
odin_ros_driver_rev1/srv/SetSensorMode "{mode: 1}"

-- /manifold/ODIN2/device0/set_sensor_mode -----服务的名字(根据ros2 service list的实际复制对应sensor mode)

-- odin_ros_driver_rev1/srv/SetSensorMode -----固定名字

-- "{mode: 1}" -----模式切换, 1: 长帧/强光 2: 短帧/弱光

四.操作示例

```
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$ ros2 service call /manifold/ODIN2/device0/set_sensor_mode odin_ros_driver_rev1/srv/SetSensorMode "{mode: 1}"
requester: making request: odin_ros_driver_rev1.srv.SetSensorMode_Request(mode=1)
response:
odin_ros_driver_rev1.srv.SetSensorMode_Response(success=True, error_code=0, message='High Peak')
xxx@xxx-MS-7E34:~/work/odin2/ros_fold$
```

ROS1

一.终端输入rosservice list查看服务列表

```
xxx@mt-ros1-env:~/work/odin2/ros_fold$ rosservice list
/manifold/ODIN2/device0/get_calibration
/manifold/ODIN2/device0/set_sensor_mode
/odin_ros_driver/6_stereo_calibra/set_parameters
/odin_ros_driver/get_loggers
/odin_ros_driver/set_logger_level
/odin_rviz/get_loggers
/odin_rviz/load_config
/odin_rviz/load_config_discarding_changes
/odin_rviz/reload_shaders
/odin_rviz/save_config
/odin_rviz/set_logger_level
/postprocess/get_loggers
/postprocess/set_logger_level
/rosout/get_loggers
/rosout/set_logger_level
xxx@mt-ros1-env:~/work/odin2/ros_fold$
```

二.终端输入source devel/setup.bash

三.服务命令例子:rosservice call /manifold/ODIN2/device0/set_sensor_mode "mode: 1"

/manifold/ODIN2/device0/set_sensor_mode -----rosservice实际的服务名字

-- "{mode: 1}" -----模式切换, 1: 长帧/强光 2: 短帧/弱光

四.操作示例

```
xxx@mt-ros1-env:~/work/odin2/ros_fold$ rosservice list
/manifold/ODIN2/device0/get_calibration
/manifold/ODIN2/device0/set_sensor_mode
/odin_ros_driver/6_stereo_calibra/set_parameters
/odin_ros_driver/get_loggers
/odin_ros_driver/set_logger_level
/odin_rviz/get_loggers
/odin_rviz/load_config
/odin_rviz/load_config_discarding_changes
/odin_rviz/reload_shaders
/odin_rviz/save_config
/odin_rviz/set_logger_level
/postprocess/get_loggers
/postprocess/set_logger_level
/rosout/get_loggers
/rosout/set_logger_level
xxx@mt-ros1-env:~/work/odin2/ros_fold$ source devel/setup.bash
xxx@mt-ros1-env:~/work/odin2/ros_fold$ rosservice call /manifold/ODIN2/device0/set_sensor_mode
Usage: rosservice call /service [args...]

rosservice: error: Please specify service arguments
xxx@mt-ros1-env:~/work/odin2/ros_fold$ rosservice call /manifold/ODIN2/device0/set_sensor_mode "mode: 1"
success: True
error_code: 0
message: "High Peak"
xxx@mt-ros1-env:~/work/odin2/ros_fold$ rosservice call /manifold/ODIN2/device0/set_sensor_mode "mode: 2"
success: True
error_code: 0
message: "Low Peak"
xxx@mt-ros1-env:~/work/odin2/ros_fold$
```